



MASTER IN ASTROPHYSICS
AND SPACE SCIENCE



TOR VERGATA
UNIVERSITÀ DEGLI STUDI DI ROMA

Optimisation & Monte Carlo Sampling

INTRODUCTION TO NUMERICAL METHODS FOR ASTROPHYSICS

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November 27, 2025

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Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

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- *Without Prior Knowledge*: out of n independent measurements the outcome A occurs k times, defining

$$P(A) = \lim_{n \rightarrow \infty} \frac{k}{n}$$

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Bayes' Theorem

$$P(A|B) = \frac{P(B|A) P(A)}{P(B)}$$

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■ *Variance* or second raw moment: $\sigma^2 = E[(X - E[X])^2] = \int_{-\infty}^{+\infty} (x - E[x])^2 f(x) dx$

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- *Arbitrary Distributions:* obtained from uniformly distributed numbers by various methods

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$$I_{\nu}(T, \beta) = B_{\nu}(T) (1 - e^{-\tau_{\nu}(\beta)}) \approx B_{\nu}(T) \tau_{\nu}(\beta) = B_{\nu}(T) \Sigma \kappa_{\nu}(\beta) = \frac{2h\Sigma\kappa_0}{c^2\nu_0^{\beta}} \frac{\nu^{3+\beta}}{\exp\left(\frac{h\nu}{kT}\right) - 1}$$

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 - suppose we have no prior information on the parameter distribution

Example

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■ *Likelihood:*

$$L(I_v | v, \sigma_v, \theta) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi\sigma_{v,i}^2}} \exp\left(-\frac{(I_{v,i} - I_v(\theta))^2}{2\sigma_{v,i}^2}\right)$$

Example

■ *Likelihood:*

$$\log L(I_v | v, \sigma_v, \theta) = -\frac{N}{2} \log(2\pi) - \sum_{i=1}^N \sigma_{v,i} - \frac{1}{2} \sum_{i=1}^N \frac{(I_{v,i} - I_v(\theta))^2}{\sigma_{v,i}^2}$$

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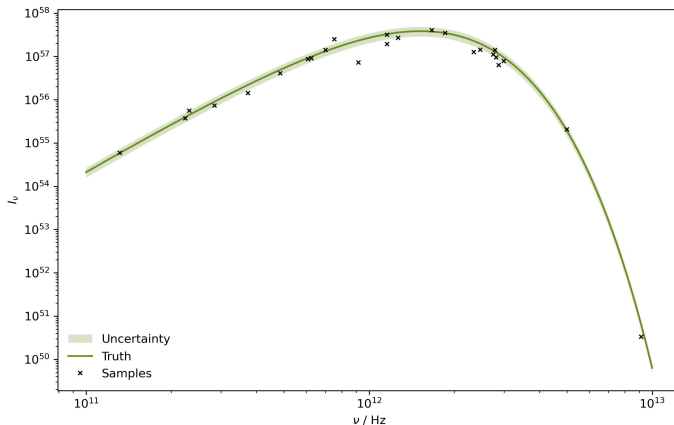
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- *Objective:* minimise the $\chi^2(\theta)$ measure to maximise the posterior probability

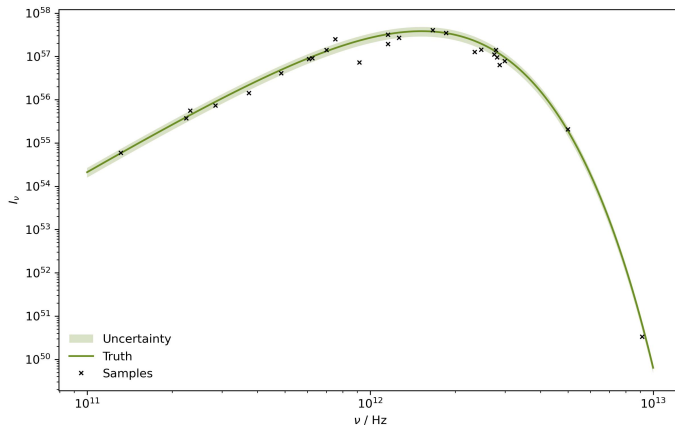
Measurements

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How do we proceed from this dataset to maximise the likelihood for our nonlinear model?



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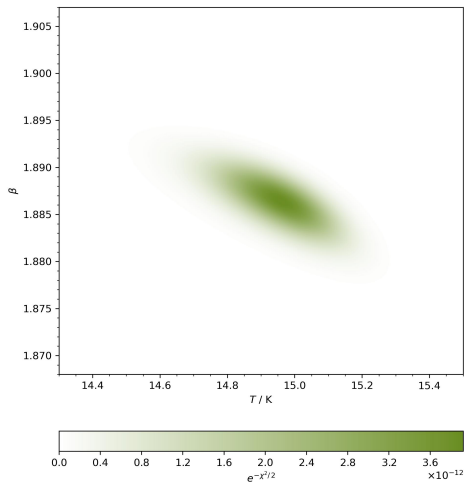
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- *Problem*: as D increases we have N^D gridpoints, and with more complex parameter distributions, this procedure quickly becomes intractable due to computational limitations

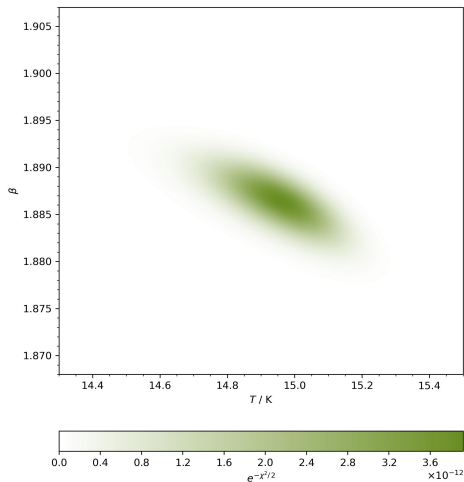
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Can we think of more elegant ways to explore even very high dimensional parameter spaces?



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- *Interactive Visualisations*: check out Ben Frederickson¹ and Andy Casey² to see some nice examples

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 - improve by considering last step and optimizing step range

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 1. initialize at some x_0
 2. update current point with $x_{k+1} = x_k - \alpha \nabla f(x_k)$
 3. iterate until convergence $|x_{k+1}/x_k - 1| < \epsilon$ or maximum step count is reached

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 - improve by performing line search for α at every step

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 - typical settings are $\tau = \kappa = 0.5$ (L. Armijo, *Pac. J. Math.* 1966)

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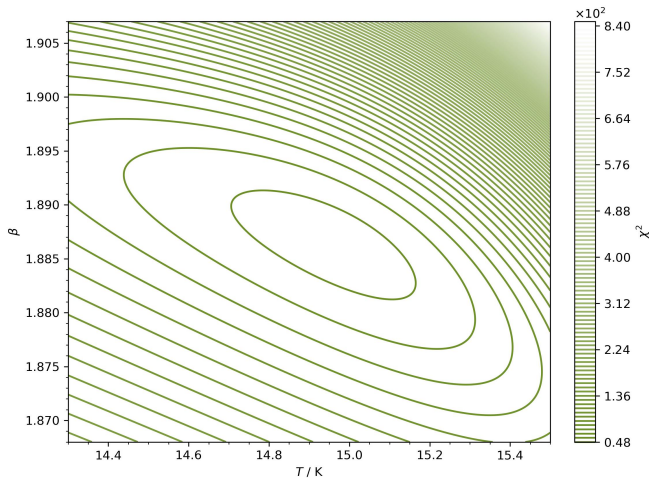
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- *Usage*: found in many modern libraries, suited to specialised use cases like nonlinear least squares, problems with parameter bounds or arbitrary constraints

Application

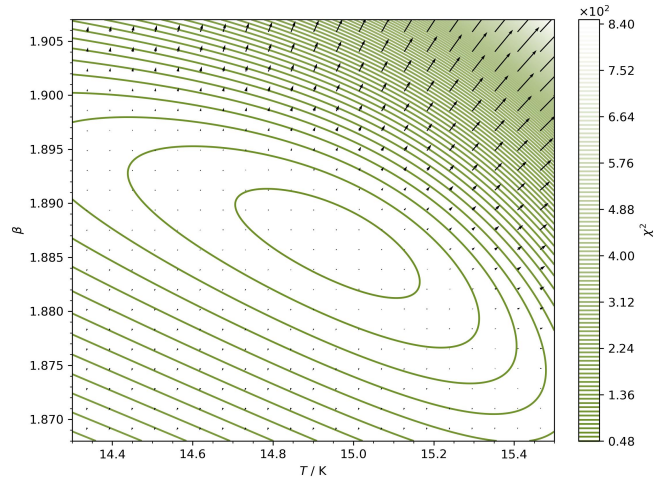
Application

- *Example:* minimise $\chi^2(\theta)$ using the gradient descent algorithm



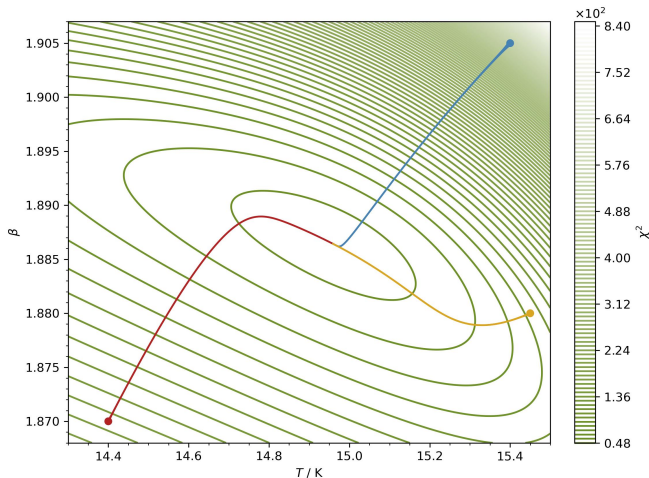
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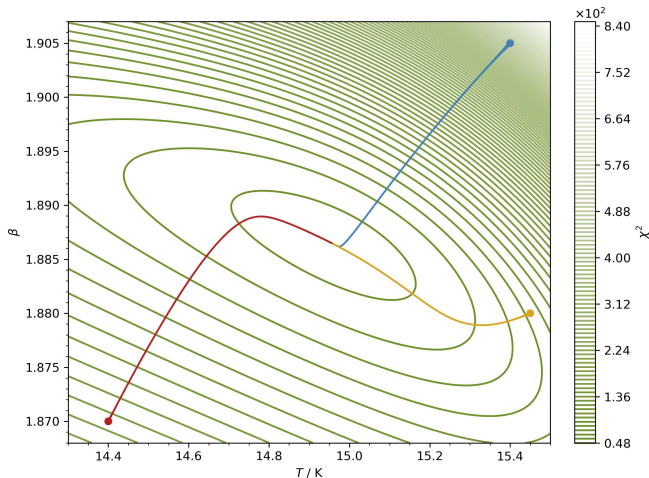
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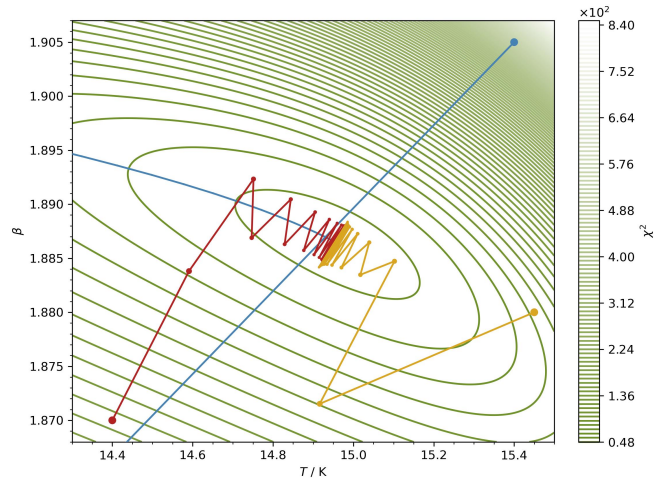
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- *Example:* minimise $\chi^2(\theta)$ using the gradient descent algorithm
- *Fixed Rate:* $\alpha = 10^{-9}$
 - very smooth, but does not converge, step size too small



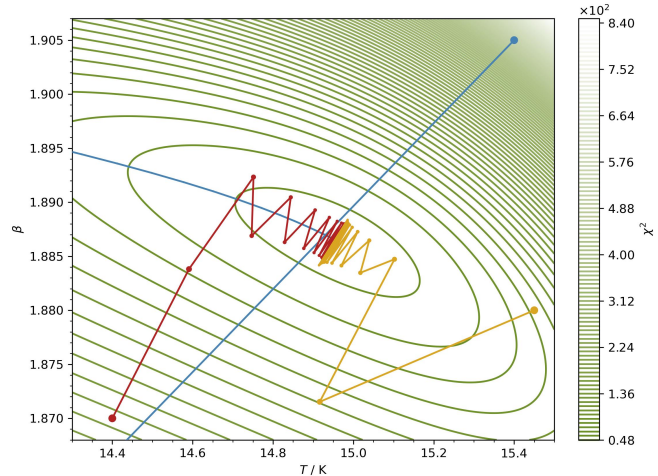
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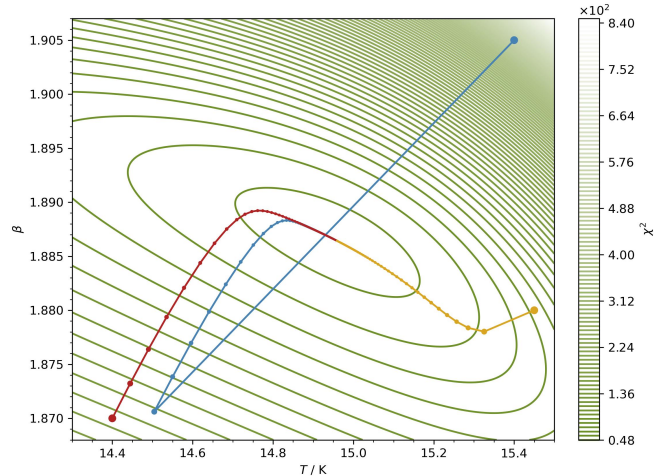
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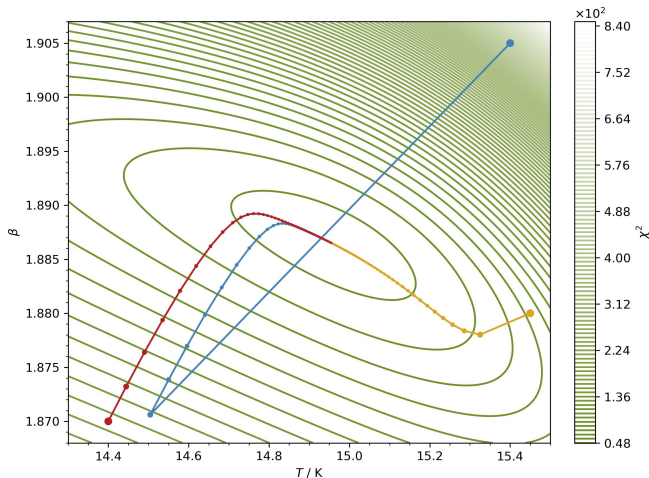
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- Example: minimise $\chi^2(\theta)$ using the gradient descent algorithm
- Fixed Rate: $\alpha = 7 \cdot 10^{-7}$



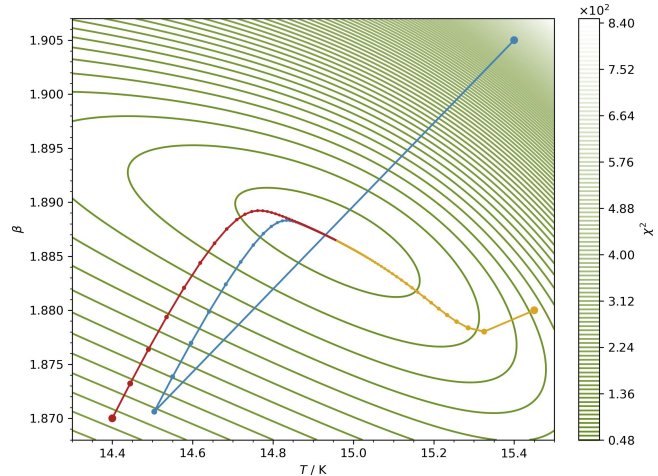
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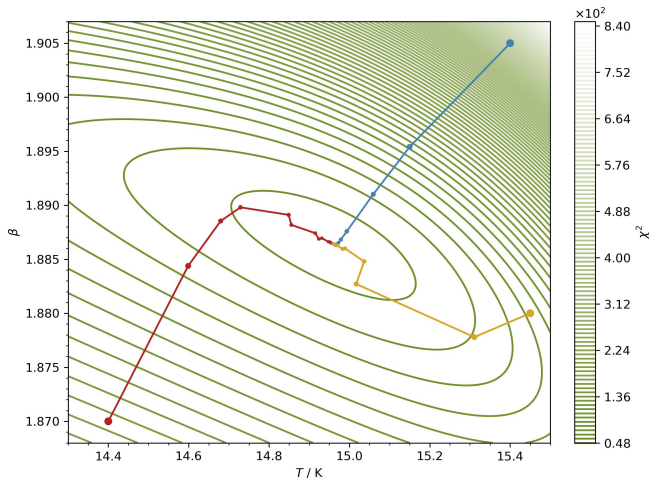
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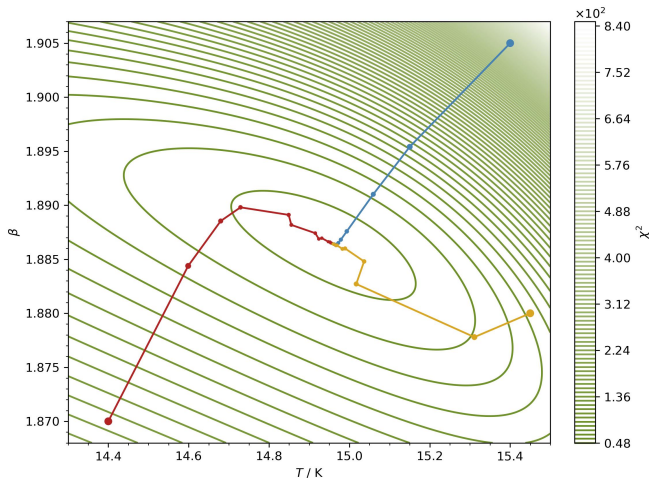
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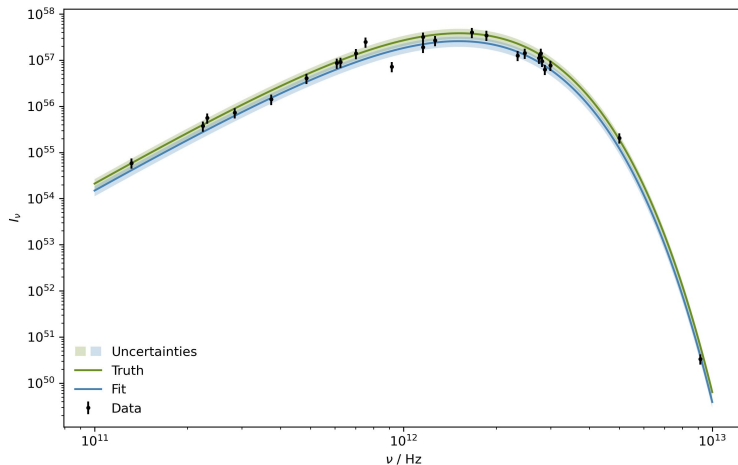


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Result



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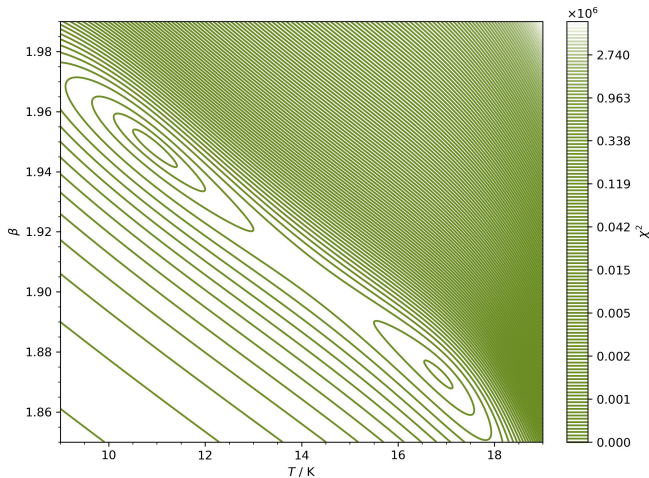
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Limitations & Curses

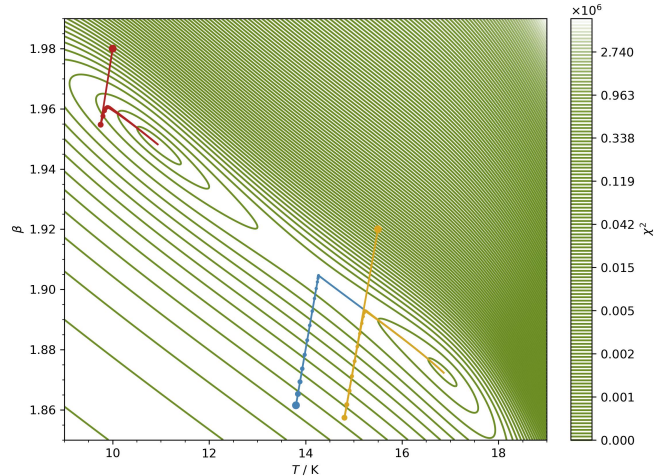
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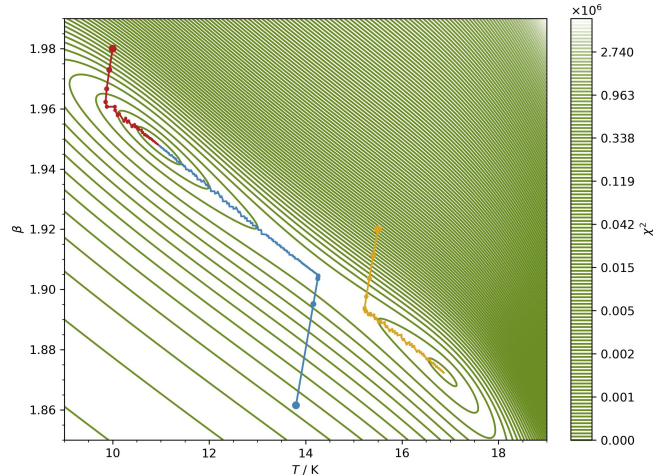
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 - Line Search Gradient Descent:
 $T_{660} = 10.93 \text{ K}$ & $\beta_{660} = 1.948$



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 - simulate complex measurement processes and quantify the total convolution

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- then the average behaviour of this single process reproduces that over the entire ensemble

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- realistic cases, especially when priors are included, are generally not symmetric or unbounded, therefore benefitting from the inbuilt bias correction to ensure correct sampling of the target distribution

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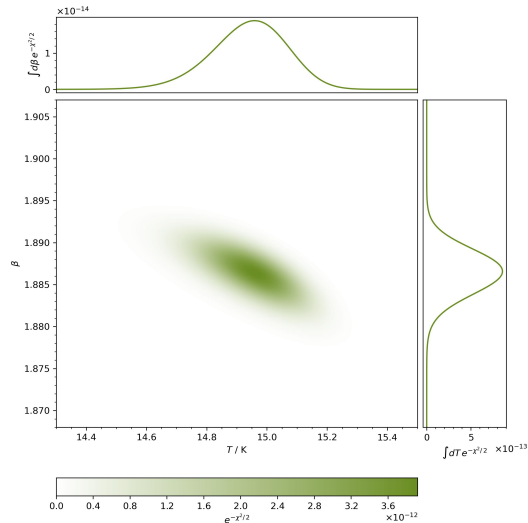
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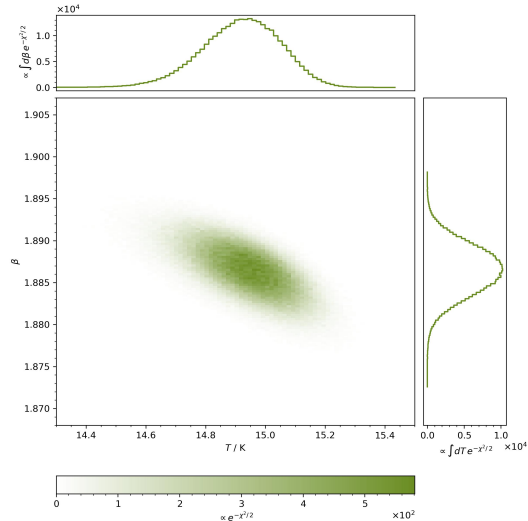
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- efficient for many dimensions with very high acceptance ratios

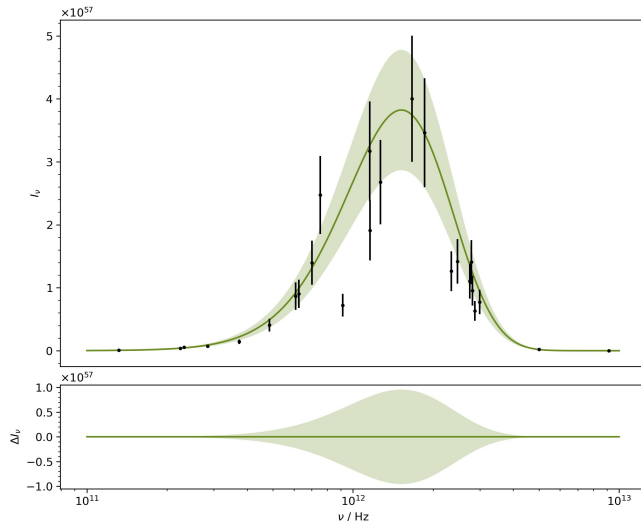
Posterior



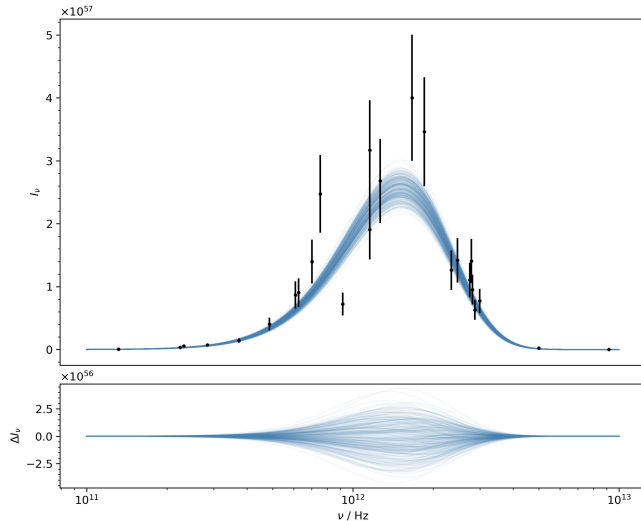
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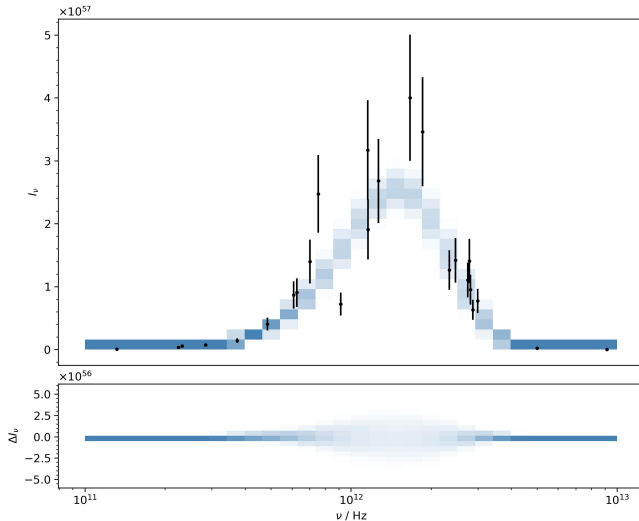
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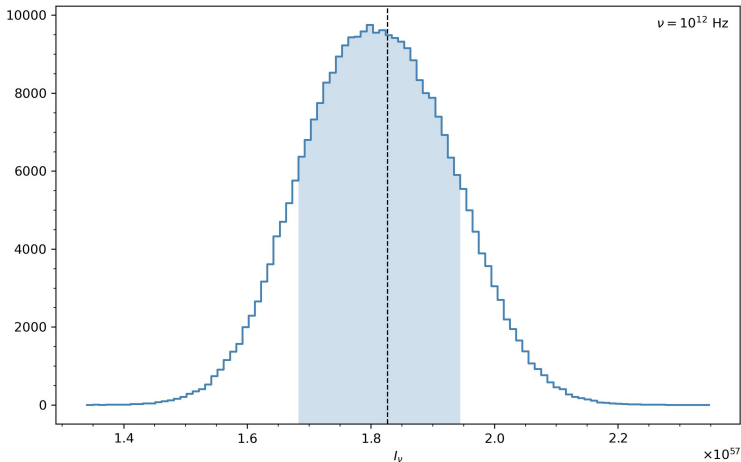
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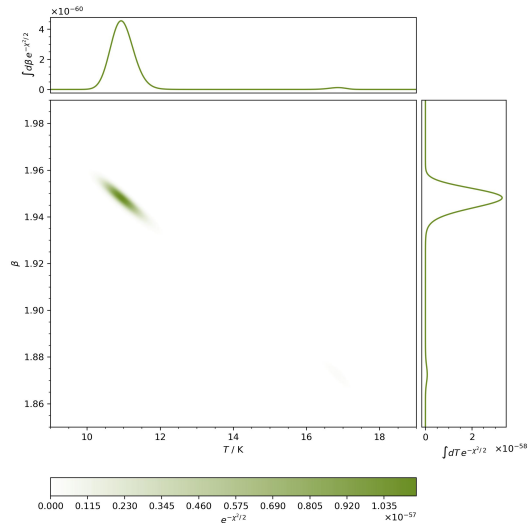
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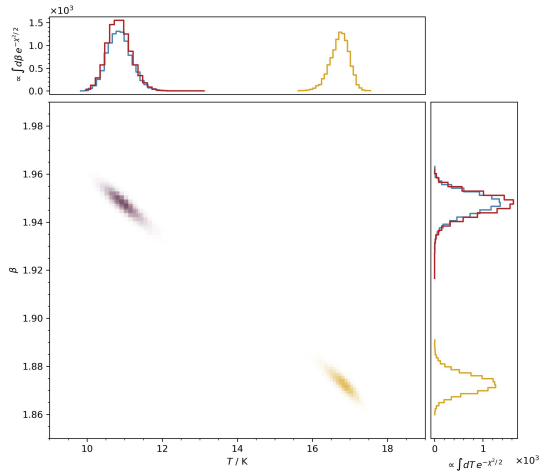
Error Propagation



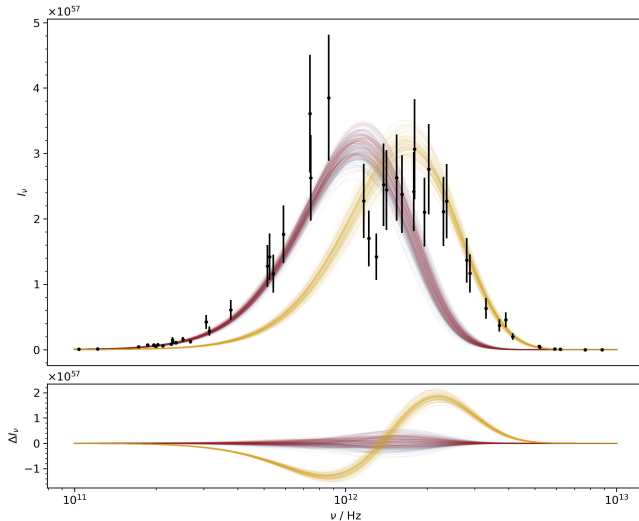
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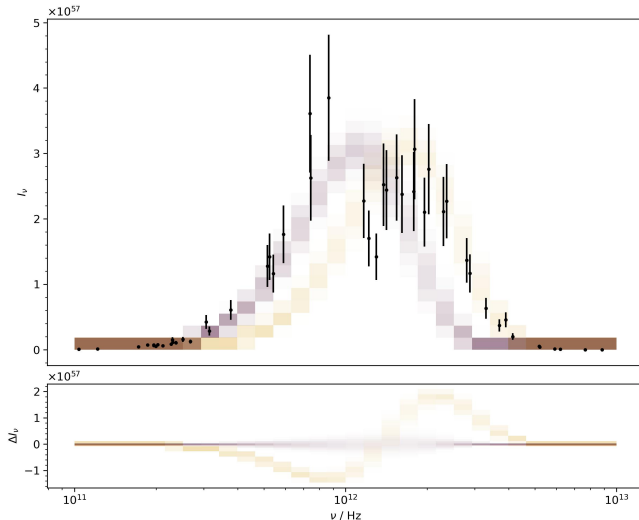
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■ Definition: $\rho_t = \frac{\text{Cov}(X_k, X_{k+t})}{\sqrt{\text{Var}(X_k)\text{Var}(X_{k+t})}} = \frac{E[(X_k - E[X_k])(X_{k+t} - E[X_{k+t}])]}{\sqrt{E[(X_k - E[X_k])^2]E[(X_{k+t} - E[X_{k+t}])^2]}}$

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Autocorrelation

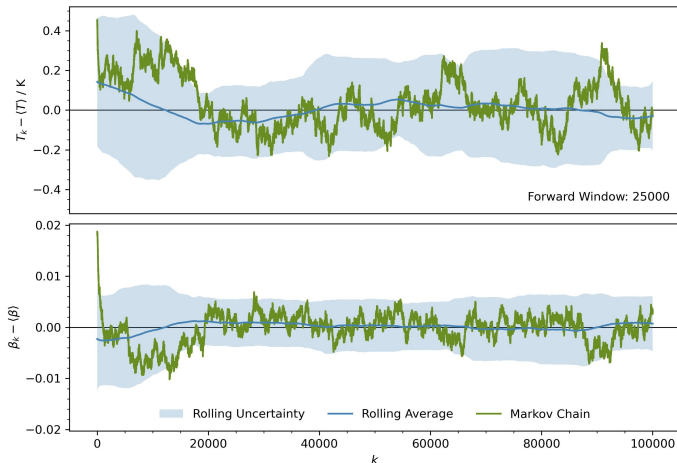
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Application

Application

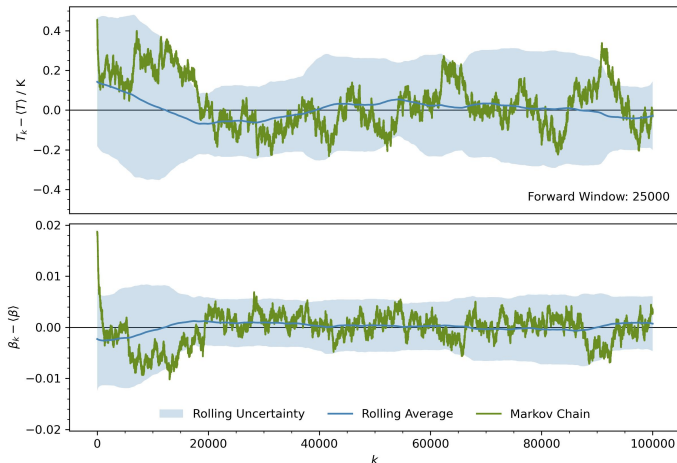
- Metropolis Algorithm: $x_j \sim N(x_i, h)$



Application

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■ $h = 10^{-4}$ & acc = 97.72%

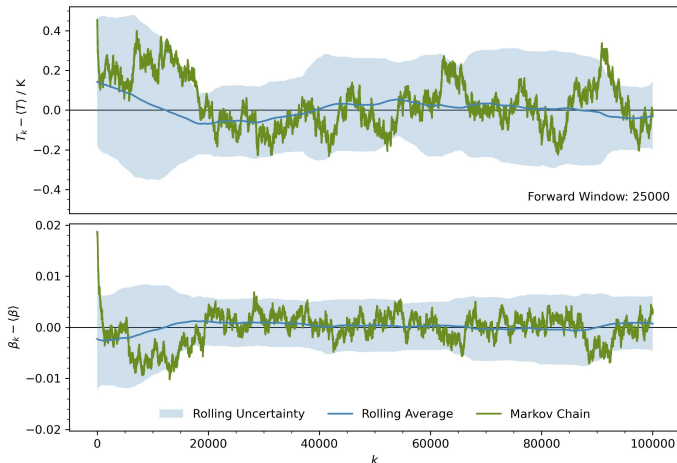


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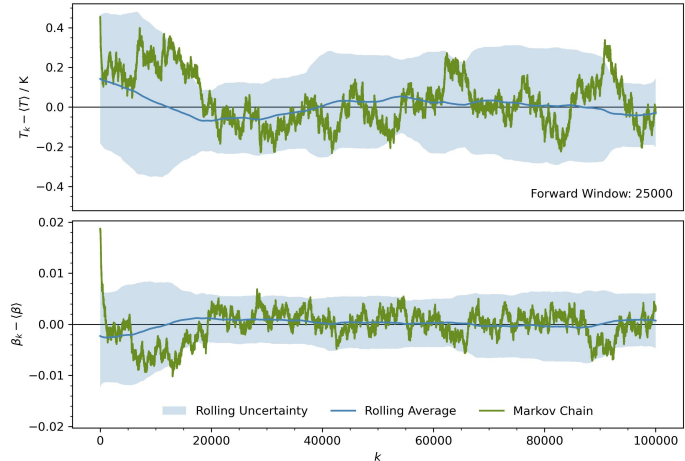
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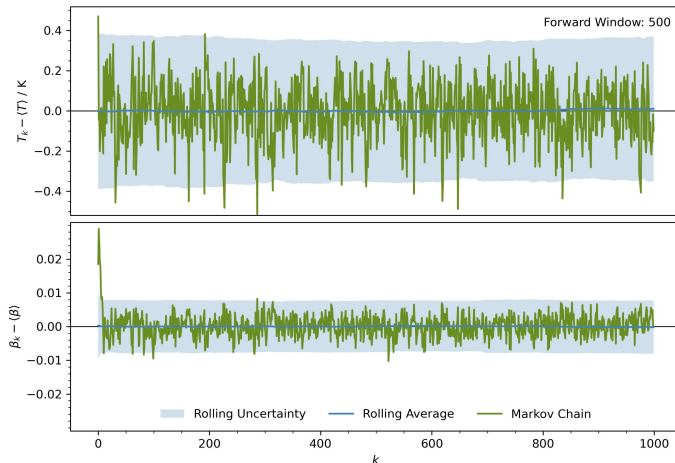
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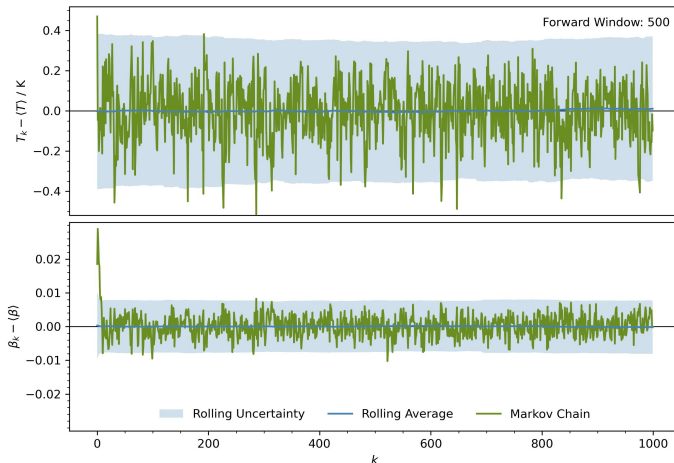
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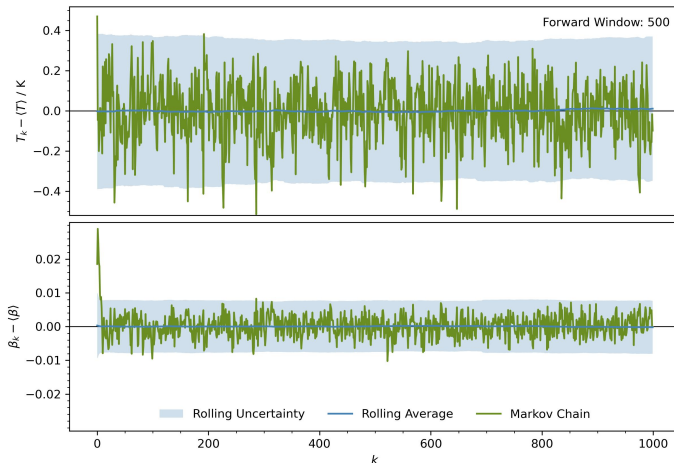


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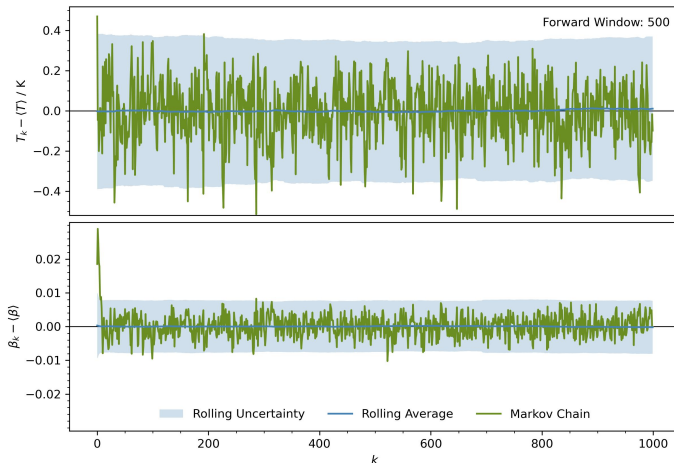
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Application

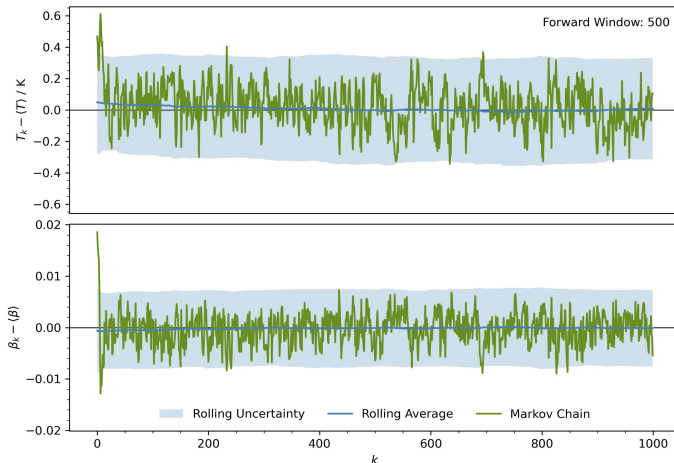
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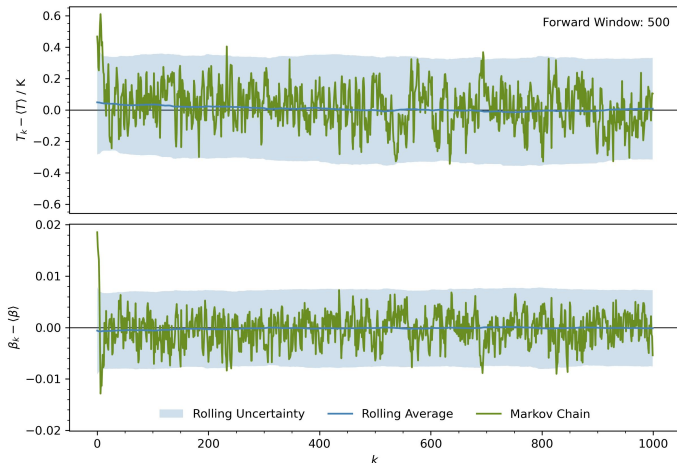
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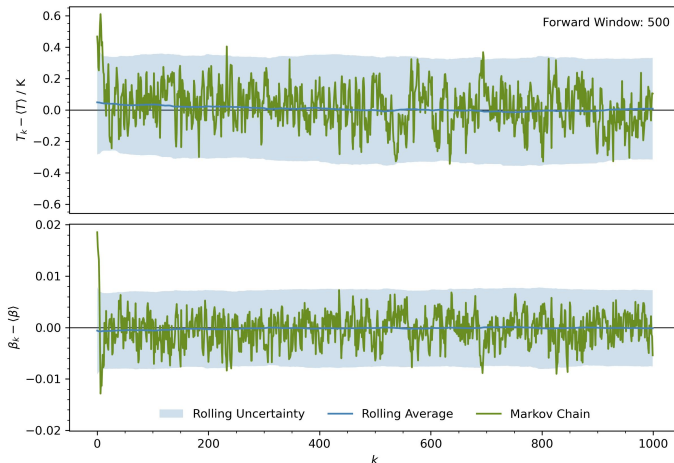


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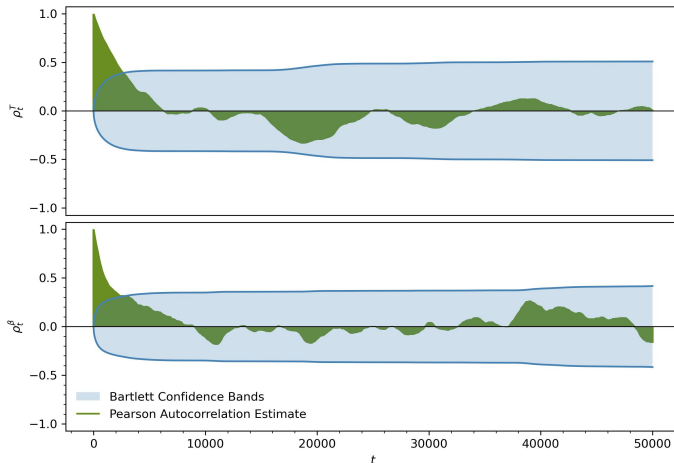
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seem good



Application

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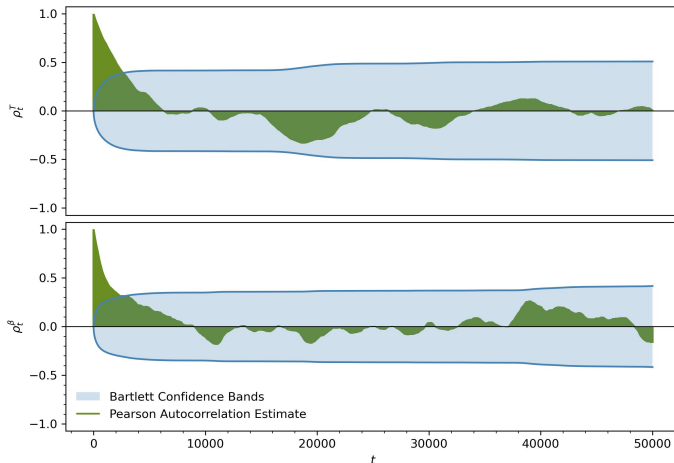
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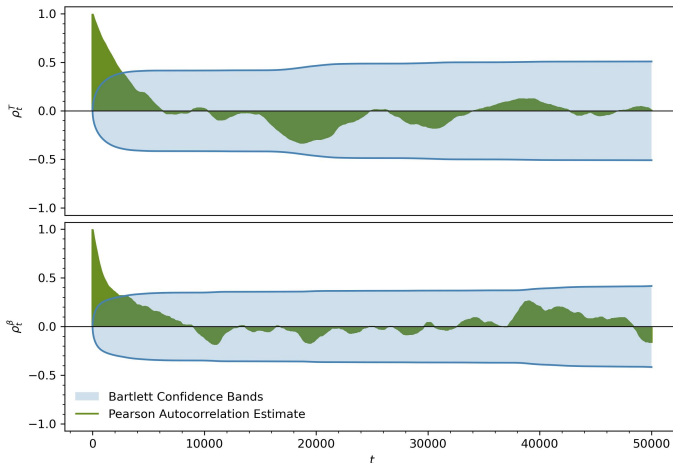


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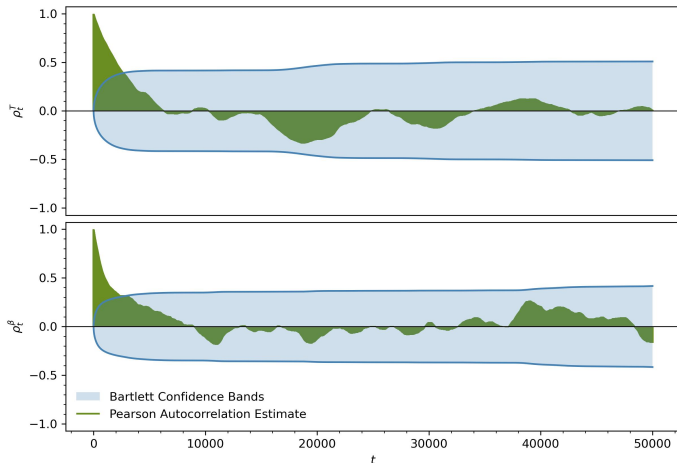
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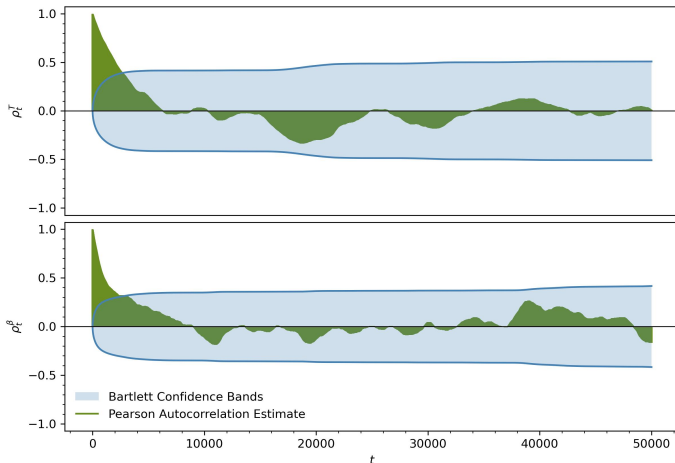
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Application

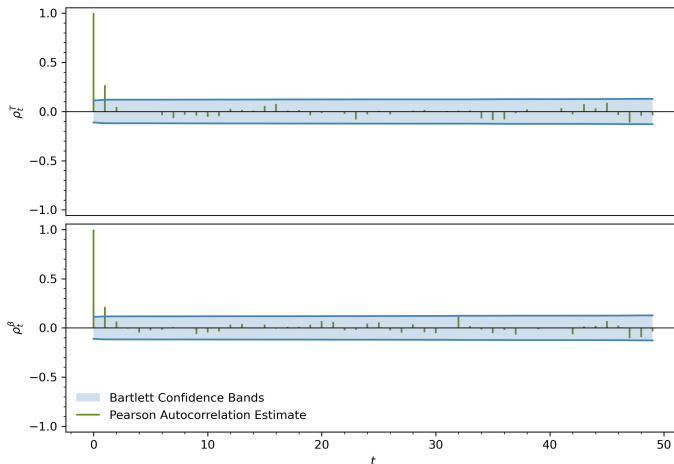
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- $l \approx 3000$



Application

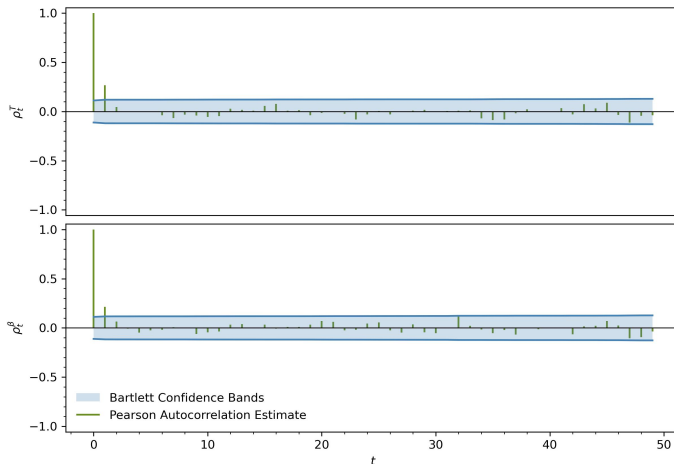
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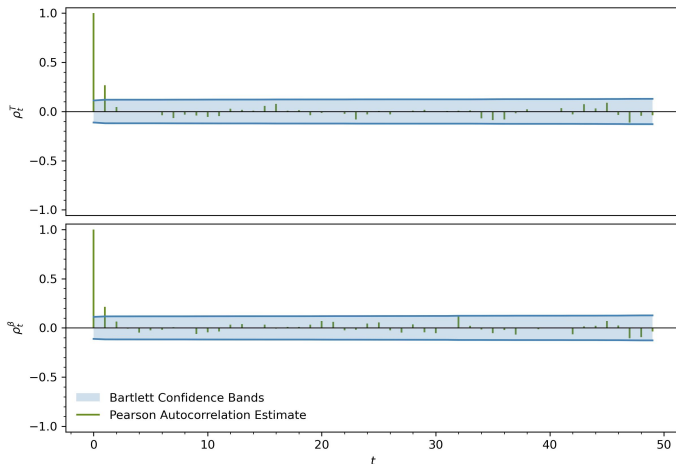


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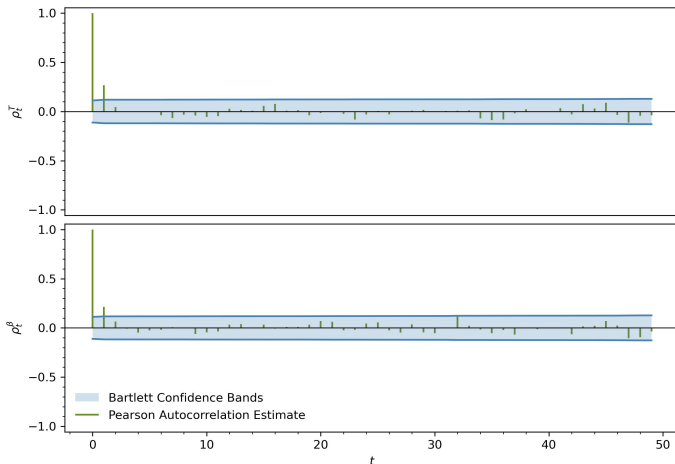
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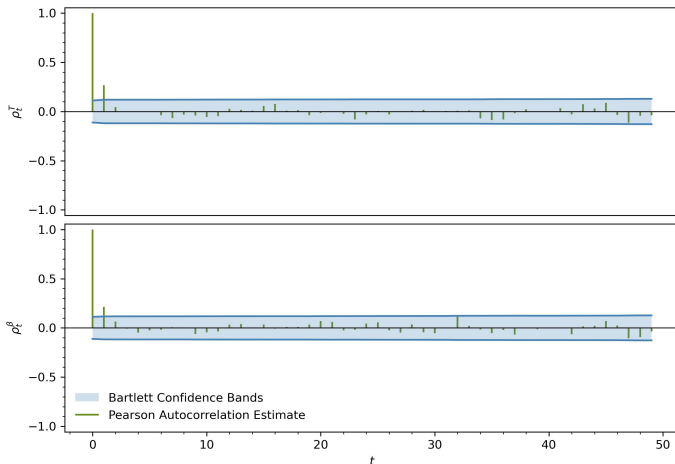
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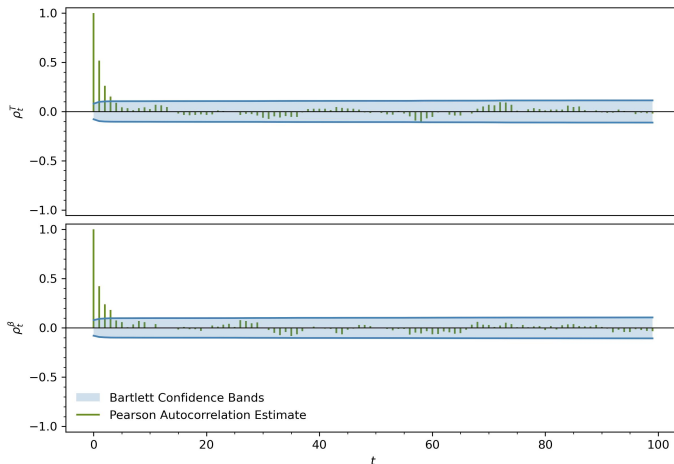
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- $l = 2$



Application

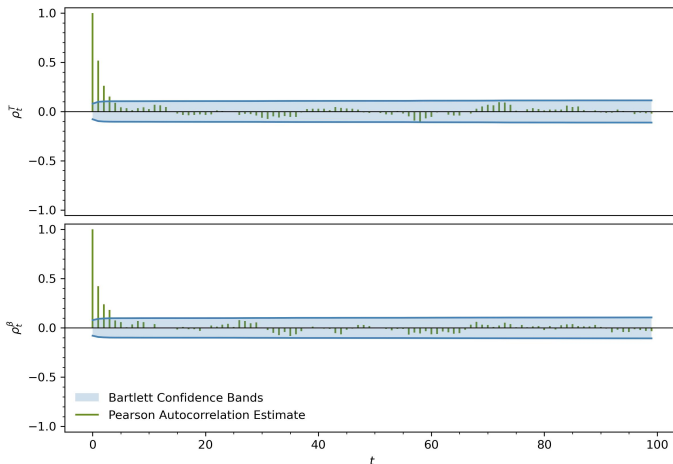
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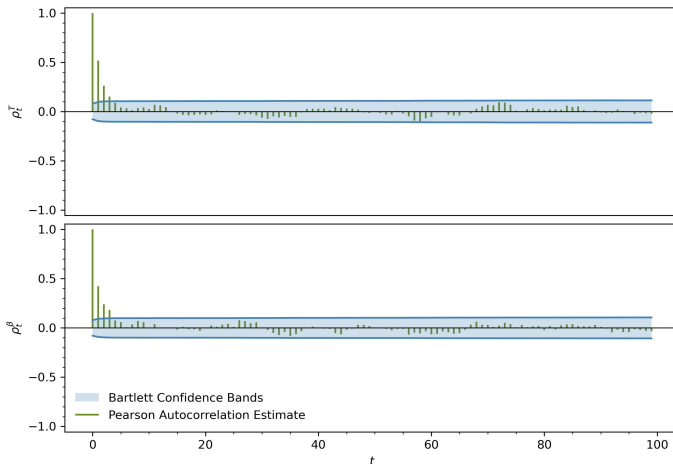


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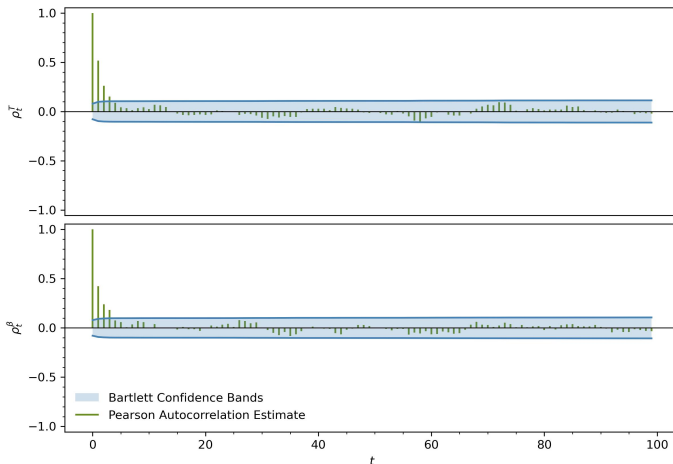
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- $l = 4$



Overview

Probability Refresher

Definitions & Notation

Random Numbers

Optimisation

Common Methods

Problems

Markov Chain Monte Carlo

Popular Algorithms

Tuning & Diagnosing

Alternatives

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 - *Importance Sampling*: weight samples from simple density to match target

Thank You!

- A. Casey, *Data Analysis and Machine Learning*³ 2022
- M. Linhoff & W. Rhode, *Statistical Methods of Data Analysis* 2023
- *emcee documentation*⁴ 2025

Resources on GitHub: [here](#)